

# Willian Vieira PhD

Data scientist and ML engineer with a PhD in quantitative ecology. I build reproducible machine learning pipelines on noisy, real-world data — from Bayesian and ML modelling to the production infrastructure that makes models traceable, monitored, and safe to deploy.

## DATA SCIENCE & ENGINEER EXPERIENCE

2025  
|  
2024  
(1 yr 9 m)

- **Data Analyst**  
**Habitat**, Montreal, Canada  
Architected end-to-end R/Python ML pipelines to process and model large-scale, noisy real-world data for production analytics | Built metadata-driven ETL and provenance workflows to make datasets, models, and analyses traceable, reproducible, and reliable | Designed cloud data infrastructure on Azure for streaming heterogeneous data and optimizing feature retrieval for ML training and inference | Pushed ML models into production, supporting client-facing products with validation checks and monitoring | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, and DevOps best practices

2024  
|  
2017

- **PhD Research**  
**Integrative Ecology Lab**, Sherbrooke, Canada  
Built Bayesian hierarchical models to forecast forest dynamics from noisy, biased, continental-scale climate and field-inventory datasets | Developed reproducible HPC workflows () to run thousands of simulations in parallel for model evaluation, uncertainty propagation, and large-scale inference | Built end-to-end data pipelines to ingest, clean, harmonize, and model heterogeneous datasets across North America | Developed open-source software libraries (, ) for spatial demographic modelling with version control, automated testing, and reproducible workflows | Published a **technical methods book** documenting the full computational pipeline, along with one peer-reviewed **publication** and two preprints (, )

2022  
|  
2020  
(part-time)

- **Biostatistician**  
**Environment and Climate Change Canada** - Quebec, Canada  
Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible **data pipeline** with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

## EDUCATION

2024  
|  
2017

- **PhD, Ecology**  
**Université de Sherbrooke** - Sherbrooke, Canada  
 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016  
|  
2015

- **Masters 2, Agroecology and Resource Management**  
**Bordeaux Sciences Agro**, Bordeaux, France  
 *Modelling the dispersion of weed species in agricultural landscapes*

2015  
|  
2010

- **BSc in Agronomy**  
**Universidade Federal de Santa Catarina**, Florianópolis, Brazil

## CONTACT

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 [w.vieiraw@gmail.com](mailto:w.vieiraw@gmail.com)  
 [LinkedIn/WillVieira](#)  
 [GitHub/WillVieira](#)  
 [Publications](#)

## SKILLS

**Machine Learning & Statistics:**  
Bayesian inference · Hierarchical models · Model evaluation · Error analysis · Uncertainty quantification · Forecasting · Feature engineering · Calibration

**Programming & Database:**  
Python · R · SQL · PyTorch · Stan · DuckDB · Bash

**Data Engineering & Cloud:**  
ETL/ELT · Production ML pipelines · Docker · Azure · AWS · Apache Arrow · HPC

**DevOps & Reproducibility:**  
Git · CI/CD · Unix · Unit testing · Reproducible pipelines · Automated validation · Make · Jupyter

## OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught programming, statistics, and ecology to 250+ students across biology, engineering, and graduate programs · Published **5 papers & preprints** with ~100 citations · Translated stakeholder needs into technical analyses, reproducible workflows, and clear analytical outputs

## LANGUAGES

Portuguese · Native  
French · Full Professional  
English · Full Professional

CV source code hosted on   
Last updated: 2026-05-15  
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